

MASTER STUDY GUIDE

Chapter 4: The Market Forces of Supply and Demand

Principles of Economics, 8th Edition | N. Gregory Mankiw

Integrated Study Notes with Bangladesh Examples, Exam Tips, and Practice Questions

PART 1: Markets and Competition

Supply and demand are the two most powerful forces in economics. They determine how much of each good is produced and at what price it is sold. To understand any economic event or policy, you must first understand how it affects supply and demand.

1.1 What Is a Market?

DEFINITION — Market

A group of buyers and sellers of a particular good or service. Buyers determine the demand side. Sellers determine the supply side. Together, they form the market.

Markets can be organized or informal. They can exist in one place or be spread across many locations.

Type of Market	Description	Bangladesh Example
Organized Market	Buyers and sellers meet at a specific place and time. Rules and prices are clear.	Dhaka Stock Exchange (DSE); commodity auctions
Informal Market	No single meeting place. Buyers choose freely from different sellers.	Karwan Bazar (vegetables and fish); local bazaars
Digital Market	Buyers and sellers interact online through a platform.	Chaldal.com (groceries); Bikroy.com (buy and sell)

Bangladesh Example — The Rice Market

Thousands of farmers grow rice across Bangladesh (supply side). Millions of families buy rice every day (demand side). Together, they form the rice market. The price of rice is not set by any one person. It is determined by the interaction of all buyers and sellers.

1.2 What Is a Competitive Market?

DEFINITION — Competitive Market

A market with so many buyers and sellers that each individual buyer or seller has a negligible impact on the market price. No single participant can control the price.

DEFINITION — Price Taker

A buyer or seller who must accept the market price as given. They are too small to influence the price.

DEFINITION — Perfect Competition

The highest level of competition, with two key conditions: (1) All goods sold are exactly identical. (2) No single buyer or seller has any influence over the market price. All participants are price takers.

Example: One rice farmer in Bangladesh sells 500 kg of rice. The total market produces millions of kilograms. That one farmer cannot raise or lower the rice price. The farmer is a price taker.

1.3 Other Market Structures

Not all markets are perfectly competitive. Mankiw introduces other structures briefly. Chapter 4 focuses on perfect competition because it is the simplest to analyze, and many lessons apply to more complex markets as well.

Market Structure	Number of Sellers	Price Control	Bangladesh Example
Perfect Competition	Many	None — all are price takers	Rice market, vegetable market
Monopoly	One	Full control over price	DESA (electricity supply in Dhaka)
Oligopoly	Few	Significant control	Mobile operators: Grameenphone, Robi, Banglalink
Monopolistic Competition	Many	Some control (unique products)	Local restaurants in Dhaka

PART 2: Demand

2.1 What Is Quantity Demanded?

DEFINITION — Demand

The relationship between the price of a good and the quantity that buyers are willing and able to purchase.

DEFINITION — Quantity Demanded

The specific amount of a good that buyers are willing AND able to purchase at a given price. Both conditions must be true: willing (desire) and able (can afford).

Important: A person who wants a BMW car (willing) but cannot afford it (not able) does not add to quantity demanded. Both conditions must be satisfied.

2.2 The Law of Demand

LAW OF DEMAND

Other things being equal, when the price of a good rises, the quantity demanded falls. When the price falls, the quantity demanded rises. Price and quantity demanded move in opposite directions (inverse relationship).

Bangladesh Example — Ilish Fish

When the price of Ilish fish is 200 Tk per kg, many families buy it regularly. When the price rises to 800 Tk per kg, fewer families buy it. During Eid, when the price rises further, only wealthy families can afford it. This is the Law of Demand in action.

2.3 The Demand Schedule

DEFINITION — Demand Schedule

A table that shows the relationship between the price of a good and the quantity demanded, holding all other factors constant.

Example — Catherine's demand for ice cream cones per month:

Price per Cone	Quantity Demanded (cones/month)
\$0.00 (Free)	12
\$0.50	10

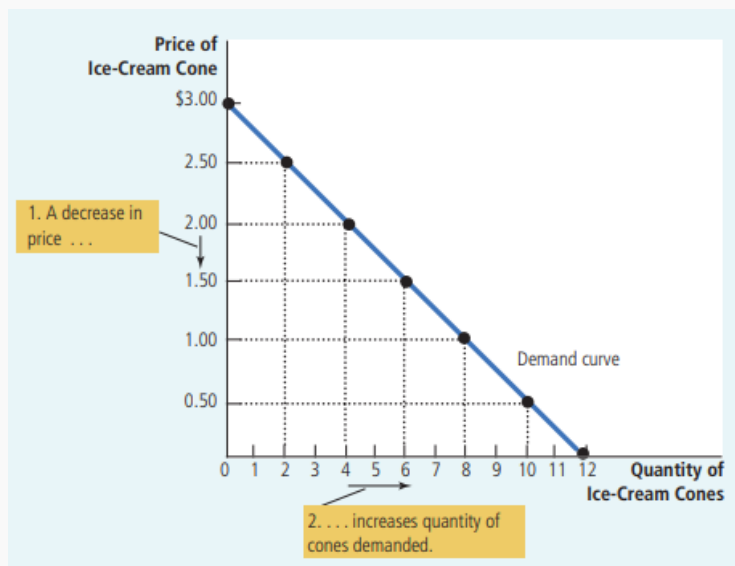
Price per Cone	Quantity Demanded (cones/month)
\$1.00	8
\$1.50	6
\$2.00	4
\$2.50	2
\$3.00	0

As price increases from \$0.00 to \$3.00, the quantity demanded falls from 12 to 0. This confirms the Law of Demand.

2.4 The Demand Curve

DEFINITION — Demand Curve

A graph that shows the relationship between the price of a good and the quantity demanded. Price (P) is on the vertical axis (Y-axis). Quantity (Q) is on the horizontal axis (X-axis). The curve slopes downward.



2.5 The Demand Equation

FORMULA — Linear Demand Equation

$$Q_d = a - bP$$

Where: Qd = Quantity demanded |
P = Price |
a = Quantity demanded when price is zero |
b = Responsiveness of demand to price (slope coefficient).
The negative sign confirms the inverse relationship.

Worked example: $Q_d = 20 - 3P$

Price (P)	Calculation	Quantity Demanded (Qd)
P = 1	$20 - 3(1) = 17$	17 units
P = 2	$20 - 3(2) = 14$	14 units
P = 3	$20 - 3(3) = 11$	11 units

2.6 Slope of the Demand Curve

FORMULA — Slope of Demand Curve

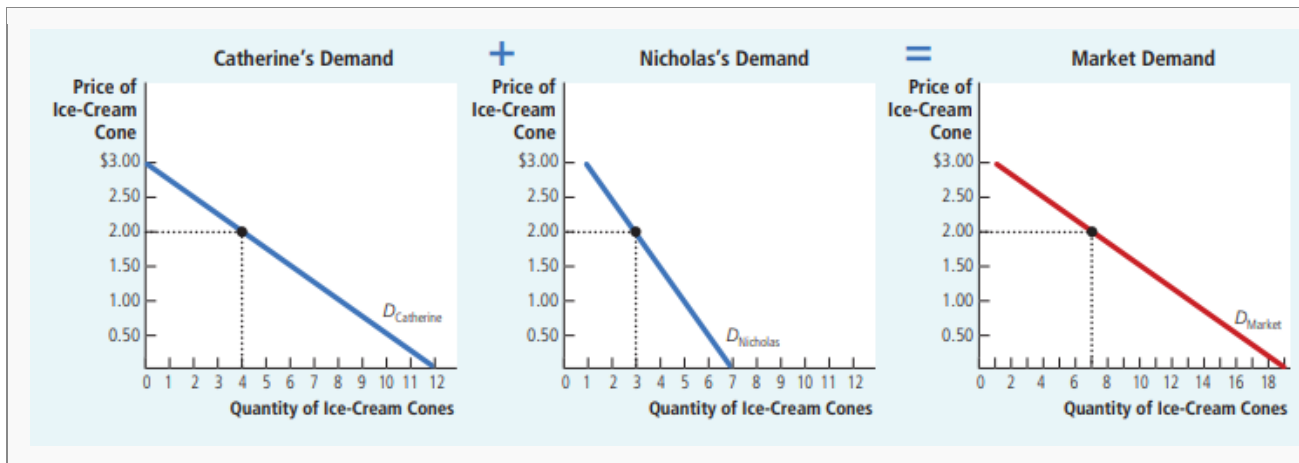
Slope = $\Delta P \div \Delta Q_d$ = (Change in Price) $\div$ (Change in Quantity Demanded) Using points (P=2, Qd=14) and (P=3, Qd=11): Slope = $(2 - 3) \div (14 - 11) = -1 \div 3 = -1/3$ The slope is negative. This confirms the demand curve slopes downward. For every 1-unit rise in price, quantity demanded falls by 3 units.

2.7 Individual Demand vs. Market Demand

Individual demand shows how much one person buys at each price. Market demand is the total demand of all buyers combined. To find market demand, add the quantities demanded by all buyers at each price level.

Example — Market demand for ice cream (Catherine + Nicholas):

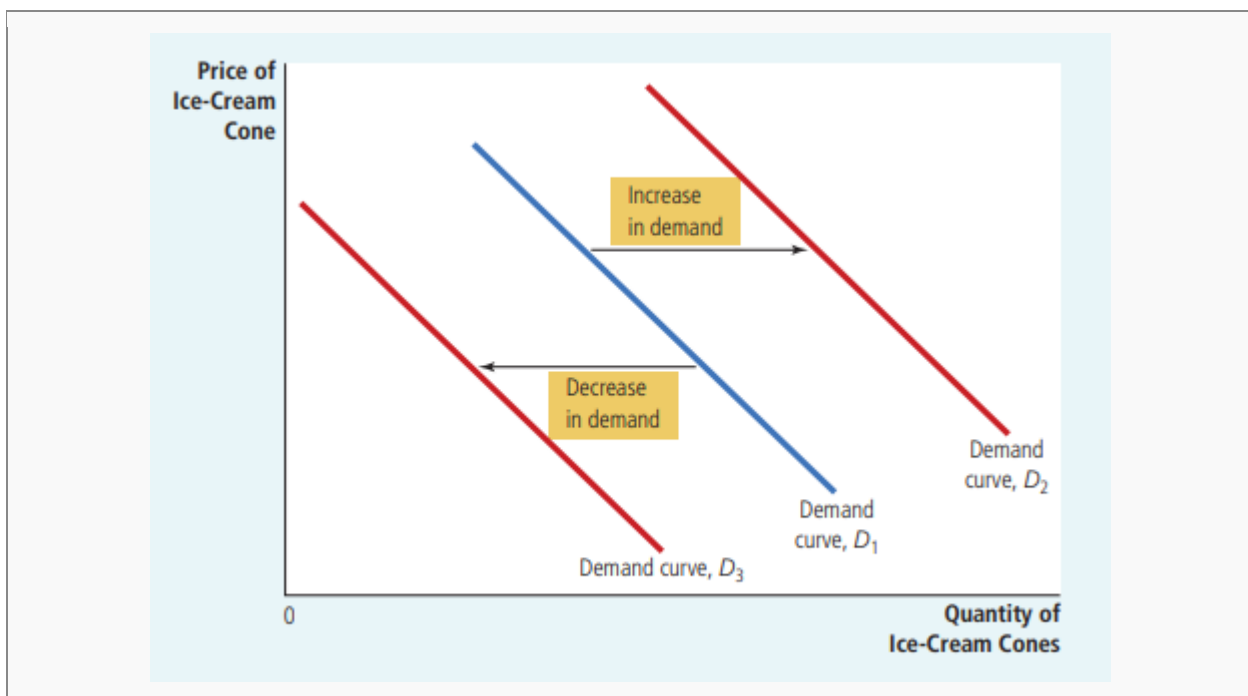
Price	Catherine	Nicholas	Market Total
\$0.00	12	7	19
\$1.00	8	5	13
\$2.00	4	3	7
\$3.00	0	1	1



2.8 Shifts in the Demand Curve

The demand curve shifts when something other than price changes. The entire curve moves to a new position.

- Shift to the RIGHT = Increase in demand (buyers want MORE at every price).
- Shift to the LEFT = Decrease in demand (buyers want LESS at every price).



Factors That Shift the Demand Curve (Determinants of Demand)

Factor	Direction of Shift	Bangladesh Example
Income rises — Normal Good	Demand INCREASES → Curve shifts RIGHT	Higher salary → buy more chicken and fish
Income rises — Inferior Good	Demand DECREASES → Curve shifts LEFT	Higher salary → buy fewer bus tickets, switch to Uber
Price of Substitute rises	Demand INCREASES → Curve shifts RIGHT	Hilsa fish price rises → demand for Rohu fish rises
Price of Complement rises	Demand DECREASES → Curve shifts LEFT	Bread price rises → demand for butter falls
Tastes improve	Demand INCREASES → Curve shifts RIGHT	Doctor recommends vegetables → more vegetable demand
Positive Expectations	Demand INCREASES → Curve shifts RIGHT	Ramadan approaching → people buy rice early
Number of Buyers increases	Demand INCREASES → Curve shifts RIGHT	Population growth → more rice demand in Bangladesh

2.9 Movement Along Curve vs. Shift of Curve

CRITICAL RULE — Movement vs. Shift (Demand)

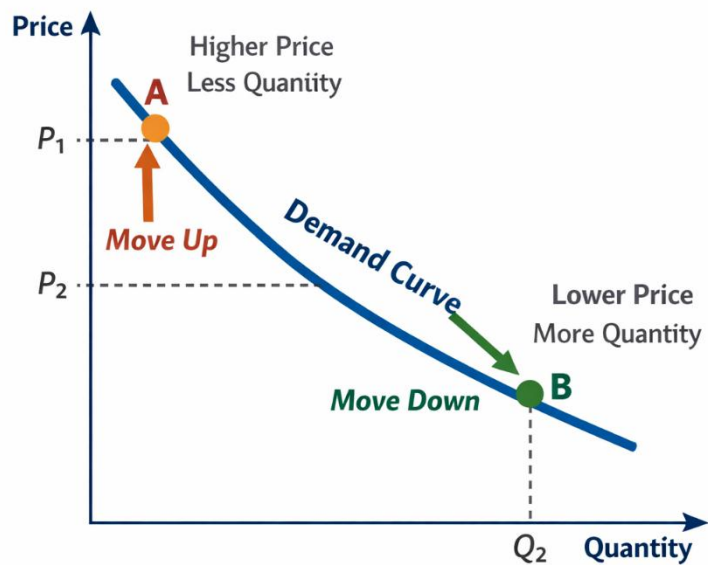
Movement **ALONG** the demand curve: Caused **ONLY** by a change in the **PRICE** of the good itself. The demand curve stays in the same place. Shift **OF** the demand curve: Caused by a change in income, related goods prices, tastes, expectations, or number of buyers. The entire curve moves to a new position.

Concept	Cause	Called	Example
Movement ALONG curve	Price of the good changes	Change in Quantity Demanded	Tea price falls → more tea demanded (move down the curve)
SHIFT of curve	Income, tastes, related goods, etc. change	Change in Demand	Cold winter → more tea demanded at every price (curve moves right)

DEFINITION — Normal Good vs. Inferior Good

Normal Good: Demand **RISES** when income rises. Examples: Rice, fish, chicken. Standard behavior. Inferior Good: Demand **FALLS** when income rises. Examples: Bus rides, cheap soap. When income rises, people switch to better alternatives. Note: 'Inferior' does not mean the product is bad. It only describes how demand responds to income.

Movement Along the Demand Curve



DEFINITION — Substitutes vs. Complements

Substitutes: Two goods that can replace each other. If the price of Good A rises, demand for Good B rises. Example: Hilsa fish and Rohu fish; rice and bread. **Complements:** Two goods used together. If the price of Good A rises, demand for Good B falls. Example: Tea and sugar; car and petrol.

PART 3: Supply

3.1 What Is Quantity Supplied?

DEFINITION — Supply

The relationship between the price of a good and the quantity that sellers are willing and able to sell.

DEFINITION — Quantity Supplied

The specific amount of a good that sellers are willing AND able to sell at a given price.

3.2 The Law of Supply

LAW OF SUPPLY

Other things being equal, when the price of a good rises, the quantity supplied rises. When the price falls, the quantity supplied falls. Price and quantity supplied move in the same direction (direct relationship).

Why does the supply curve slope upward? When a price is high, selling is more profitable. Sellers work longer hours, hire more workers, and produce more. When a price is low, production is less profitable. Sellers produce less — some may stop entirely.

Bangladesh Example — Garment Industry

When the global price of t-shirts rises, Bangladeshi garment factories increase production. They run extra shifts, hire more workers, and buy more fabric. Quantity supplied increases. This is the Law of Supply in action.

3.3 The Supply Schedule and Supply Curve

DEFINITION — Supply Schedule

A table that shows the relationship between the price of a good and the quantity supplied, holding all other factors constant.

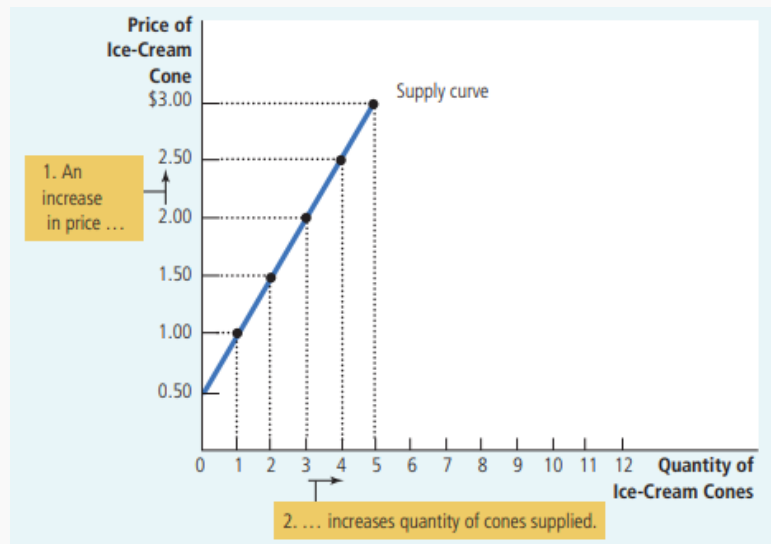
Example — Ben's supply of ice cream cones per month:

Price per Cone	Quantity Supplied (cones/month)
\$0.00	0
\$0.50	0
\$1.00	1

Price per Cone	Quantity Supplied (cones/month)
\$1.50	2
\$2.00	3
\$2.50	4
\$3.00	5

DEFINITION — Supply Curve

A graph that shows the relationship between price and quantity supplied. Price (P) is on the vertical axis. Quantity (Q) is on the horizontal axis. The curve slopes upward (positive slope).



3.4 Individual Supply vs. Market Supply

Individual supply shows how much one seller supplies at each price. Market supply is the total from all sellers combined.

Price	Ben	Jerry	Market Total
\$1.00	1	0	1
\$1.50	2	2	4
\$2.00	3	4	7
\$2.50	4	6	10

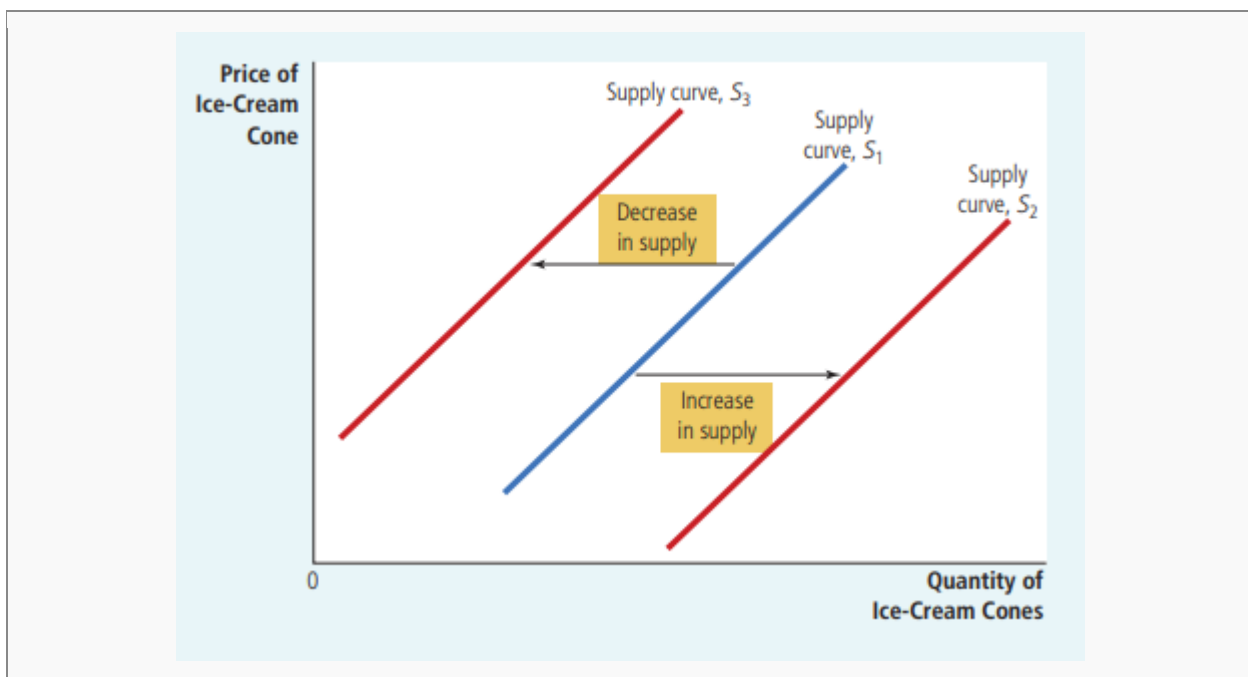
Price	Ben	Jerry	Market Total
\$3.00	5	8	13

The market supply curve is found by adding all individual supply curves horizontally.

3.5 Shifts in the Supply Curve

The supply curve shifts when something other than price changes.

- Shift to the RIGHT = Increase in supply (sellers want to sell MORE at every price).
- Shift to the LEFT = Decrease in supply (sellers want to sell LESS at every price).



Factors That Shift the Supply Curve (Determinants of Supply)

Factor	Direction of Shift	Bangladesh Example
Input Prices rise	Supply DECREASES → Curve shifts LEFT	Gas price rises → transport costs rise → fewer vegetables reach market
Input Prices fall	Supply INCREASES → Curve shifts RIGHT	Cheaper fabric → garment factories produce more shirts
Technology improves	Supply INCREASES → Curve shifts RIGHT	New rice harvesting machine → more rice production at lower cost

Factor	Direction of Shift	Bangladesh Example
Sellers expect higher future prices	Supply DECREASES today → Curve shifts LEFT	Sellers store goods now to sell at higher prices later
Number of Sellers increases	Supply INCREASES → Curve shifts RIGHT	More garment factories open → more clothing produced

3.6 Movement Along Curve vs. Shift of Curve (Supply)

CRITICAL RULE — Movement vs. Shift (Supply)

Movement **ALONG** the supply curve: Caused **ONLY** by a change in the **PRICE** of the good itself. Shift **OF** the supply curve: Caused by a change in input prices, technology, expectations, or number of sellers.

3.7 Summary Table — Supply Curve Shifts

Variable	What Happens to Supply Curve
Price of the good itself changes	Movement ALONG the curve — not a shift
Input prices rise	Supply decreases → Curve shifts LEFT
Input prices fall	Supply increases → Curve shifts RIGHT
Technology improves	Supply increases → Curve shifts RIGHT
Sellers expect higher future prices	Supply decreases today → Curve shifts LEFT
Number of sellers increases	Supply increases → Curve shifts RIGHT

PART 4: Supply and Demand Together — Equilibrium

4.1 What Is Equilibrium?

DEFINITION — Equilibrium

The point where the supply curve and the demand curve intersect. At this point, quantity supplied equals quantity demanded. The market is balanced.

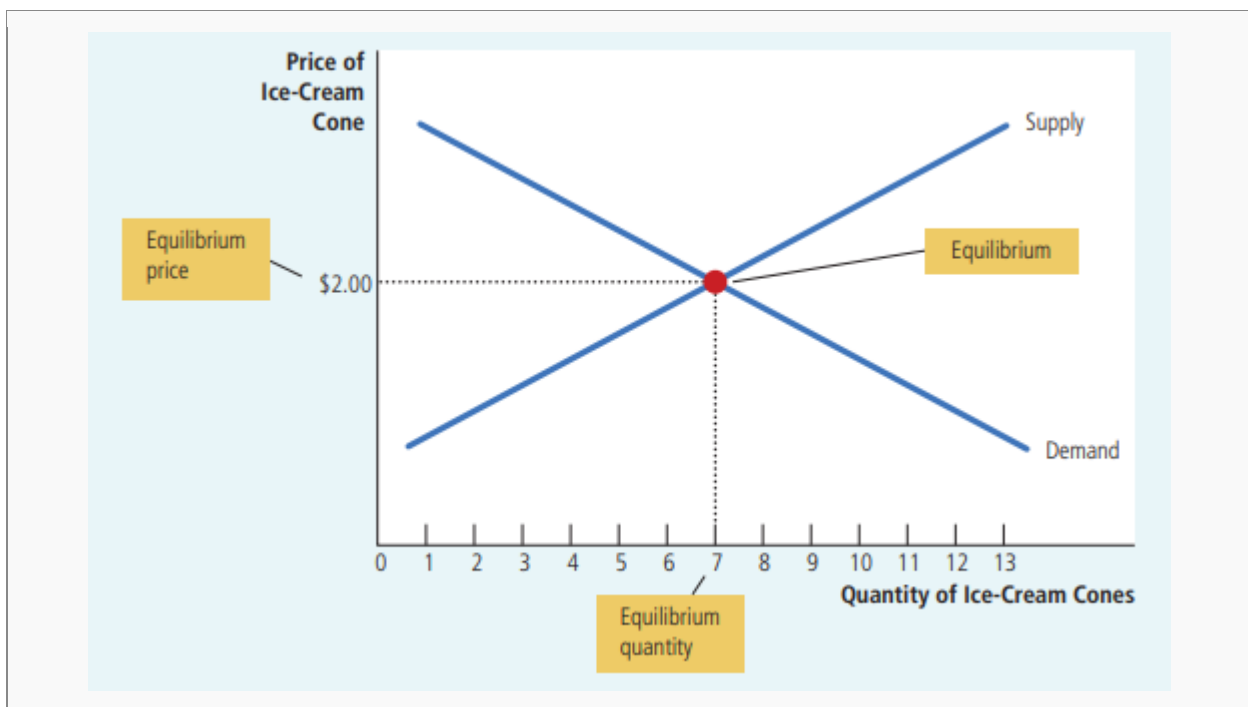
DEFINITION — Equilibrium Price

The price at which quantity supplied equals quantity demanded. It is also called the market-clearing price.

DEFINITION — Equilibrium Quantity

The quantity of a good bought and sold at the equilibrium price.

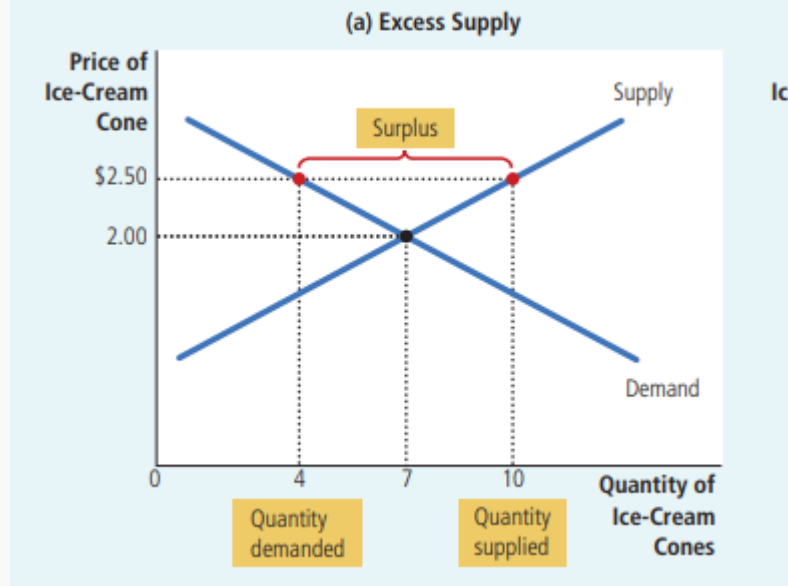
Example: In the textbook ice cream market, equilibrium price = \$2.00 per cone and equilibrium quantity = 7 cones. At this price, sellers want to sell exactly 7 cones, and buyers want to buy exactly 7 cones. The market is balanced.



4.2 Surplus and Shortage

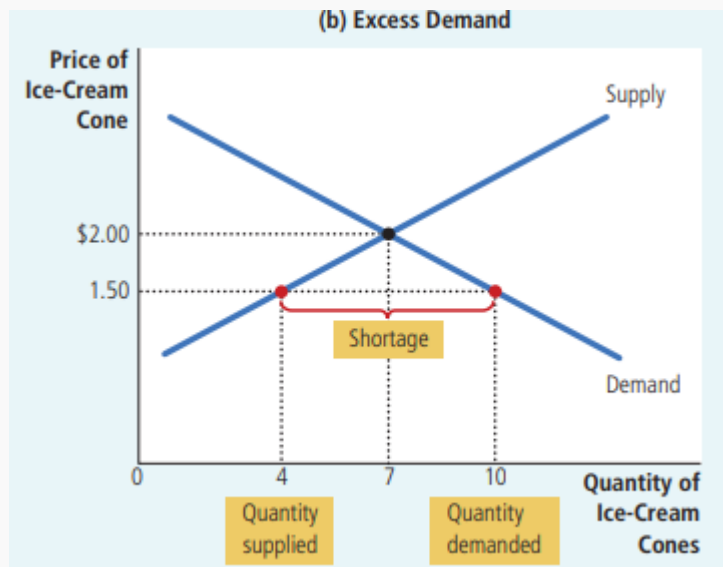
DEFINITION — Surplus (Excess Supply)

When the market price is ABOVE the equilibrium price. Quantity supplied exceeds quantity demanded. Sellers have more goods than buyers want. Sellers respond by lowering prices. The market returns to equilibrium. Example: Price = \$2.50. Supply = 10 cones. Demand = 4 cones. Surplus = 6 cones.



DEFINITION — Shortage (Excess Demand)

When the market price is BELOW the equilibrium price. Quantity demanded exceeds quantity supplied. Buyers compete for limited goods. Sellers raise prices. The market returns to equilibrium. Example: Price = \$1.50. Demand = 10 cones. Supply = 4 cones. Shortage = 6 cones.



LAW OF SUPPLY AND DEMAND

The price of any good adjusts automatically to bring the quantity supplied and the quantity demanded into balance. Markets naturally move toward equilibrium over time.

Situation	What Happens	Result
Price > Equilibrium (Surplus)	Sellers have unsold goods. They cut prices.	Price falls back toward equilibrium.
Price < Equilibrium (Shortage)	Buyers compete for scarce goods. Sellers raise prices.	Price rises back toward equilibrium.
Price = Equilibrium	Market clears. No pressure on price.	Price remains stable.

Bangladesh Example — Eid Cattle Market

Before Eid, demand for cattle rises sharply. There are not enough cattle to meet demand (shortage). Prices rise. After Eid, many unsold cattle remain (surplus). Prices fall. The market naturally moves toward equilibrium in both cases.

PART 5: Analyzing Changes in Equilibrium

5.1 The Three-Step Method

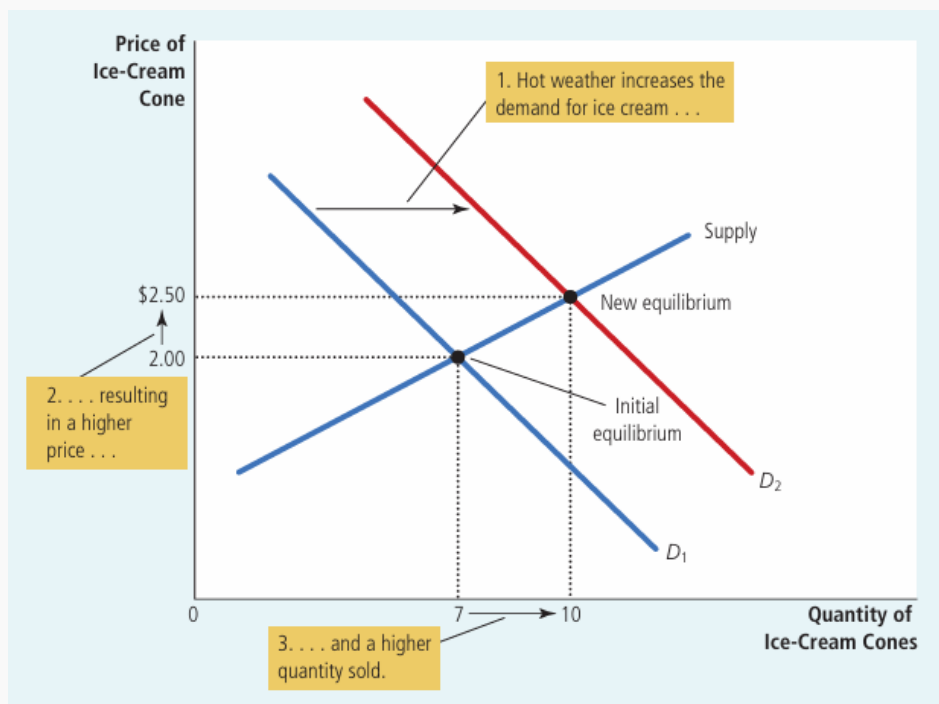
THE 3-STEP METHOD — Analyzing Any Market Change

Step 1: Decide whether the event shifts the SUPPLY curve, the DEMAND curve, or BOTH.
Step 2: Decide which DIRECTION the curve shifts (shift RIGHT = increase; shift LEFT = decrease). Step 3: Draw the new supply-and-demand diagram. Compare the new equilibrium to the old equilibrium. Note how price and quantity changed.

5.2 Worked Example 1: Demand Increases (Hot Weather and Ice Cream)

Event: A very hot summer occurs. How does this affect the ice cream market?

1. Step 1: Hot weather affects the DEMAND curve. It changes how much people want to buy at any given price. The supply curve is not affected.
2. Step 2: Hot weather makes people want more ice cream. Demand INCREASES. Demand curve shifts to the RIGHT.
3. Step 3: At the old price of \$2.00, there is now excess demand (shortage). Sellers raise prices. New equilibrium: price rises from \$2.00 to \$2.50. Quantity rises from 7 to 10 cones.



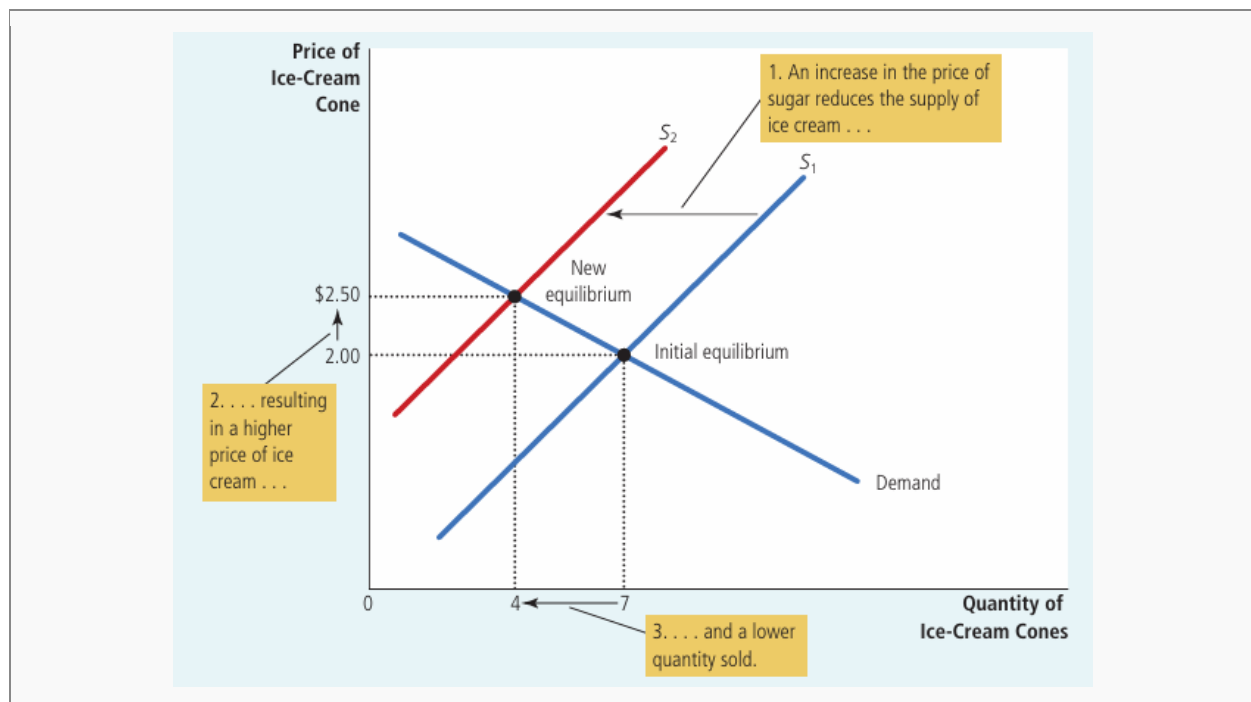
KEY RESULT

An increase in demand raises BOTH the equilibrium price AND the equilibrium quantity.

5.3 Worked Example 2: Supply Decreases (Sugar Price Rises)

Event: A hurricane destroys sugarcane crops. Sugar prices rise. How does this affect the ice cream market?

- Step 1: Higher sugar prices affect the SUPPLY curve. Sugar is an input for making ice cream. Higher input costs reduce supply. The demand curve is not affected.
- Step 2: Higher costs mean sellers supply LESS at every price. Supply DECREASES. Supply curve shifts to the LEFT.
- Step 3: At the old price of \$2.00, there is now excess demand. Sellers raise prices. New equilibrium: price rises from \$2.00 to \$2.50. Quantity falls from 7 to 4 cones.



KEY RESULT

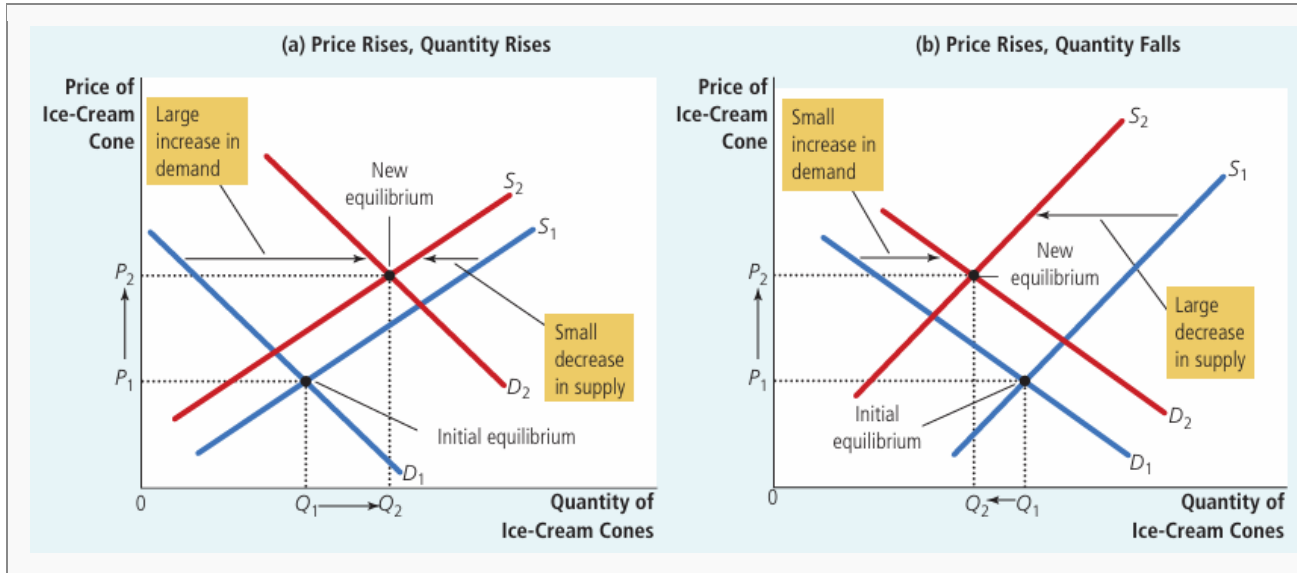
A decrease in supply raises the equilibrium price and lowers the equilibrium quantity.

5.4 Worked Example 3: Both Curves Shift Simultaneously

Event: A hot summer (increases demand) AND a hurricane that raises sugar prices (decreases supply) occur at the same time.

- Step 1: BOTH curves shift.
- Step 2: Demand curve shifts RIGHT (hot weather). Supply curve shifts LEFT (higher sugar costs).
- Step 3: Both shifts push price UPWARD. The direction of quantity change depends on which shift is larger.

Scenario	Effect on Price	Effect on Quantity
Demand rises more than supply falls	Price rises	Quantity rises
Supply falls more than demand rises	Price rises	Quantity falls
Both shifts are equal	Price rises	Quantity unchanged



KEY RESULT

When both curves shift, the direction of the price change is usually certain. The direction of the quantity change is ambiguous — it depends on the relative size of each shift.

5.5 Complete Summary Table — All Possible Curve Shifts

Event	Curve Shift	Price Effect	Quantity Effect
Demand increases	D shifts RIGHT	Rises	Rises
Demand decreases	D shifts LEFT	Falls	Falls
Supply increases	S shifts RIGHT	Falls	Rises
Supply decreases	S shifts LEFT	Rises	Falls
D increases + S increases	Both RIGHT	Ambiguous	Rises
D decreases + S decreases	Both LEFT	Ambiguous	Falls
D increases + S decreases	D right, S left	Rises	Ambiguous

Event	Curve Shift	Price Effect	Quantity Effect
D decreases + S increases	D left, S right	Falls	Ambiguous

5.6 Master Outcomes Table (Rows = Demand; Columns = Supply)

	Supply Unchanged	Supply Increases	Supply Decreases
Demand Unchanged	P same, Q same	P falls, Q rises	P rises, Q falls
Demand Increases	P rises, Q rises	P ambiguous, Q rises	P rises, Q ambiguous
Demand Decreases	P falls, Q falls	P falls, Q ambiguous	P ambiguous, Q falls

PART 6: How Prices Allocate Resources

In a market economy, prices serve a very important signaling role. They guide decisions without any central authority giving orders.

Price Signal	Message to Buyers	Message to Sellers
High price	This good is scarce or valuable. Buy less.	This good is profitable. Produce more.
Low price	This good is plentiful. Buy more freely.	This good is less profitable. Produce less.

CONCEPT — Adam Smith's Invisible Hand

Adam Smith described prices as an 'invisible hand' that guides the economy. Nobody plans the economy centrally. Yet millions of individual decisions — all guided by prices — lead to an efficient allocation of resources across society. The price system is the tool the invisible hand uses.

Bangladesh Example — The IT Labor Market

When IT companies in Dhaka need more programmers, they offer higher salaries (a higher price for labor). This signal attracts more people to study computer science. The market, through price alone, guides human resources to where they are needed — without any central planner deciding.

PART 7: All Key Terms — Quick Reference Glossary

Term	Simple Definition
Market	A group of buyers and sellers of a particular good or service.
Competitive Market	A market with many buyers and sellers; price is set by the market, not by any individual.
Price Taker	A buyer or seller who accepts the market price and cannot change it.
Perfect Competition	All goods are identical; no single person can influence the price.
Monopoly	Only one seller in the market; that seller sets the price.
Oligopoly	A few large sellers who have significant influence over price.
Monopolistic Competition	Many sellers, each with a slightly different product; some price control.
Quantity Demanded	The amount buyers are willing and able to purchase at a given price.
Law of Demand	Price rises → Quantity demanded falls. An inverse relationship.
Demand Schedule	A table showing quantity demanded at each price level.
Demand Curve	A graph of the demand schedule; slopes downward.
Normal Good	Demand rises when income rises. Example: fish, chicken.
Inferior Good	Demand falls when income rises. Example: bus rides, cheap soap.
Substitutes	Two goods where a price rise in one increases demand for the other.
Complements	Two goods used together; price rise in one reduces demand for the other.
Quantity Supplied	The amount sellers are willing and able to sell at a given price.
Law of Supply	Price rises → Quantity supplied rises. A direct relationship.
Supply Schedule	A table showing quantity supplied at each price level.
Supply Curve	A graph of the supply schedule; slopes upward.
Equilibrium	The point where quantity supplied equals quantity demanded. The market is balanced.
Equilibrium Price	The price at which the market clears (quantity supplied = quantity demanded).
Equilibrium Quantity	The amount bought and sold at the equilibrium price.

Term	Simple Definition
Surplus	Quantity supplied exceeds quantity demanded; price is too high.
Shortage	Quantity demanded exceeds quantity supplied; price is too low.
Law of Supply and Demand	Price adjusts automatically until quantity supplied equals quantity demanded.
Invisible Hand	Adam Smith's concept that prices guide resources to their most valued uses without central planning.

PART 8: Key Comparisons and Distinctions

8.1 Movement Along a Curve vs. Shift of the Curve

Concept	Cause	What Moves	Graph Effect
Movement along demand curve	Price of the same good changes	One point to another on the same curve	Travel up or down the curve
Shift of demand curve	Income, tastes, related goods, expectations, or number of buyers changes	The entire demand curve	A new curve appears (left or right)
Movement along supply curve	Price of the same good changes	One point to another on the same curve	Travel up or down the curve
Shift of supply curve	Input prices, technology, expectations, or number of sellers changes	The entire supply curve	A new curve appears (left or right)

8.2 Surplus vs. Shortage

Feature	Surplus (Excess Supply)	Shortage (Excess Demand)
When does it occur?	Market price is ABOVE equilibrium	Market price is BELOW equilibrium
Who has the problem?	Sellers cannot sell everything	Buyers cannot buy everything they want
Market response	Sellers cut prices	Sellers raise prices
Final result	Price falls toward equilibrium	Price rises toward equilibrium
Numeric example	Supply(10) > Demand(4); surplus of 6	Demand(10) > Supply(4); shortage of 6

8.3 Normal Goods vs. Inferior Goods

Feature	Normal Good	Inferior Good
When income rises	Demand increases	Demand decreases
When income falls	Demand decreases	Demand increases
Bangladesh examples	Rice, fish, chicken, clothing	Bus rides, cheap soap, instant noodles

8.4 Substitutes vs. Complements

Feature	Substitutes	Complements
Definition	Goods that replace each other	Goods used together
If price of Good A rises	Demand for Good B rises	Demand for Good B falls
Bangladesh examples	Hilsa fish and Rohu fish; Rice and bread	Tea and sugar; Car and petrol
Textbook examples	Ice cream and frozen yogurt	Ice cream and hot fudge

PART 9: Practice Questions with Answers

Section A — Markets and Competition

Q1. What is a competitive market? Describe a market that is not perfectly competitive.

MODEL ANSWER

A competitive market has many buyers and sellers, with no single participant able to affect the market price. All participants are price takers. A non-competitive example: DESA (electricity in Dhaka) is a monopoly — one seller sets the price. Grameenphone, Robi, and Banglalink form an oligopoly in the mobile market — a few large sellers with significant pricing power.

Section B — Demand

Q2. Does a change in PRICE shift the demand curve or cause a movement along it?

MODEL ANSWER

A change in the PRICE of the good itself causes a MOVEMENT ALONG the demand curve. Quantity demanded changes, but demand does not shift. A change in income, tastes, related goods prices, expectations, or number of buyers causes a SHIFT of the entire demand curve.

Q3. Harry's income falls, and he buys more pumpkin juice. Is pumpkin juice a normal good or an inferior good?

MODEL ANSWER

Pumpkin juice is an INFERIOR good. When Harry's income falls, his demand for pumpkin juice increases. The demand curve for pumpkin juice shifts to the RIGHT (increases). This is the defining characteristic of an inferior good.

Section C — Supply and Equilibrium

Q4. Does a change in technology shift the supply curve or cause a movement along it?

MODEL ANSWER

A change in technology causes a SHIFT of the supply curve. Better technology lowers production costs, so sellers supply more at every price. The supply curve shifts to the RIGHT (supply increases). A change in the PRICE of the good causes a movement along the existing supply curve.

Q5. The price of tomatoes rises. What happens in the pizza market?

MODEL ANSWER

Tomatoes are an input for making pizza. Step 1: The supply of pizza is affected (supply curve shifts). Step 2: Higher tomato prices raise production costs. Supply decreases —

supply curve shifts LEFT. Step 3: At the old price, there is now excess demand. Price rises. New equilibrium: pizza price rises and pizza quantity falls.

Q6. Beer and pizza are complements. If the price of beer rises, what happens to the pizza market?

MODEL ANSWER

Beer price rises → people buy less beer → since beer and pizza are complements (consumed together), demand for pizza falls → pizza demand curve shifts LEFT → pizza equilibrium price falls → pizza equilibrium quantity falls.

Q1: What is demand? What are the determinants of demand?

Demand is the quantity of a good that buyers are WILLING and ABLE to purchase at various prices.

Determinants of Demand (factors that SHIFT the demand curve):

1. Consumer income (normal vs. inferior goods)
2. Price of related goods (substitutes and complements)
3. Tastes and preferences of buyers
4. Expectations of future price changes
5. Number of buyers in the market

Q2: What is the Law of Demand? Prove the demand curve slopes downward.

Law of Demand: When price rises, quantity demanded falls (inverse relationship).

Proof using equation $Q_d = 20 - 3P$:

$P=1: Q_d=17$ $P=2: Q_d=14$ $P=3: Q_d=11$

As P rises, Q_d falls → slope = $-1/3$ (NEGATIVE) → downward sloping.

Reason:

Income effect: higher price means buyers can afford less.

Substitution effect: buyers switch to cheaper substitutes.

Q3: Given a demand equation, draw a curve and calculate slope.

Example: $Q_d = 20 - 3P$

1. Make table: (P=1, Q_d=17), (P=2, Q_d=14), (P=3, Q_d=11)
2. Plot P on vertical axis, Q_d on horizontal axis.
3. Connect points → straight downward line.

$$\text{Slope} = \Delta P / \Delta Q_d = (2-3) / (14-11) = -1/3$$

Q4: Difference between movement along curve vs. shift in demand curve.

	Movement Along Curve	Shift of Curve
Cause	Price of the good changes	Income, tastes, related goods, expectations
Curve stays?	YES — same curve	NO — whole curve moves left or right
Name	Change in Quantity Demanded	Change in Demand

Q5: If consumer income rises and raw material price falls — what happens to equilibrium?

Consumer income rises → Demand INCREASES → D shifts RIGHT

Raw material price falls → Production cost falls → Supply INCREASES → S shifts RIGHT

Result:

Equilibrium Quantity: DEFINITELY RISES (both shifts push Q up)

Equilibrium Price: AMBIGUOUS (D increase pushes P up; S increase pushes P down)

Final price depends on which shift is larger.

Q6: Define Price Elasticity of Demand. Give the formula.

Price Elasticity of Demand (E_p) measures how sensitive Q_d is to a change in price.

Formula: $E_p = (\% \text{ Change in } Q_d) / (\% \text{ Change in } P)$

Or: $E_p = (dQ/dP) \times (P/Q)$

If $|E_p| > 1$ → Elastic (sensitive buyers)

If $|E_p| < 1 \rightarrow$ Inelastic (not sensitive buyers)

If $|E_p| = 1 \rightarrow$ Unit elastic

Q7: Given $P=10$, $Q=200$ and $P=8$, $Q=250$ — calculate price elasticity.

$$dP = 10 - 8 = 2$$

$$dQ = 200 - 250 = -50$$

$$E_p = (dQ/dP) \times (P/Q)$$

$$= (-50/2) \times (10/200)$$

$$= -25 \times 0.05$$

$$= -1.25$$

$|E_p| = 1.25 > 1 \rightarrow$ Demand is ELASTIC

PART 10: Exam Tips and Memory Aids

10.1 Top 5 Common Exam Mistakes

MISTAKE 1 — Confusing a shift with a movement along the curve

RULE: A change in the PRICE of the good causes a movement ALONG the curve. A change in any other factor (income, tastes, related goods, etc.) causes a SHIFT of the entire curve.

MISTAKE 2 — Saying 'both supply and demand increase, so price definitely rises'

RULE: If BOTH curves shift in the same direction, the price effect is AMBIGUOUS. We cannot determine the direction of the price change without knowing which shift is larger.

MISTAKE 3 — Confusing 'inferior good' with a low-quality product

RULE: An inferior good is NOT necessarily a bad product. It simply means demand FALLS when income RISES. The term describes behavior, not product quality.

MISTAKE 4 — Getting the direction of surplus and shortage wrong

RULE: Price TOO HIGH above equilibrium = SURPLUS (excess supply). Price TOO LOW below equilibrium = SHORTAGE (excess demand). Remember: surplus means sellers have too much. Shortage means buyers cannot get enough.

MISTAKE 5 — Thinking equilibrium is reached instantly

RULE: Markets move TOWARD equilibrium over time. The speed depends on how quickly prices adjust in that particular market.

10.2 Memory Tips

DEMAND

D is for Downward slope. The demand curve always slopes DOWN from left to right.

SUPPLY

S is for upSlope. The supply curve always slopes UP from left to right.

SURPLUS vs. SHORTAGE

Surplus = Supply EXCEEDS demand → too much on the shelf → sellers CUT prices.
Shortage = Short of supply → not enough for everyone → sellers RAISE prices.

NORMAL vs. INFERIOR GOOD

Normal good: NORMAL behavior — income up, buy MORE. Inferior good: When I am RICH, I do not want it anymore — income up, buy LESS.

SUBSTITUTES vs. COMPLEMENTS

SUBSTITUTES can be Substituted for each other: Hilsa price up → buy Rohu instead. COMPLEMENTS Complete each other: Tea and sugar always go together.

THE 3-STEP METHOD (for exam questions)

Step 1: Which CURVE shifts? (supply or demand or both?) Step 2: Which DIRECTION? (right = increase, left = decrease) Step 3: What happens to PRICE and QUANTITY? (compare old and new equilibrium)

10.3 Chapter 4 — Final Summary

Topic	Key Point to Remember
Market	A group of buyers (demand) and sellers (supply) of a good or service.
Perfect Competition	Many buyers and sellers; all are price takers; identical products.
Law of Demand	Price up → Quantity demanded down. Inverse relationship.
Law of Supply	Price up → Quantity supplied up. Direct relationship.
Demand Curve	Slopes downward. Shifts right = increase. Shifts left = decrease.
Supply Curve	Slopes upward. Shifts right = increase. Shifts left = decrease.
Equilibrium	Where supply = demand. Market-clearing price and quantity.
Surplus	Price too high → $Q_s > Q_d$ → sellers cut prices → return to equilibrium.
Shortage	Price too low → $Q_d > Q_s$ → sellers raise prices → return to equilibrium.
Simultaneous shifts	Two shifts: one outcome is certain; the other is ambiguous.
Prices as signals	High price = scarce/valuable. Low price = plentiful. Prices guide resources.
Invisible Hand	Prices coordinate millions of decisions without central planning.

Study Tip: Read each section of these notes. Then close the document. Write down the key points from memory. Repeat this process until you are confident in every topic.

*End of Master Study Guide — Chapter 4: The Market Forces of Supply and Demand
Principles of Economics, 8th Edition | N. Gregory Mankiw*

SUPPLEMENTARY NOTES — TOPICS FROM CLASS IMAGES

Chapter 4: The Market Forces of Supply and Demand

Principles of Economics, 8th Edition | N. Gregory Mankiw

The following topics were covered in your class images but are NOT in the original notes above.

SECTION A: Branches of Economics & Core Concepts

A.1 Micro vs. Macro Economics

Two main branches of Economics:

Microeconomics — studies individual decisions: one buyer, one firm, one market.

Macroeconomics — studies the whole economy: GDP, inflation, unemployment.

Branch	Focus Area	Bangladesh Example
Microeconomics	Individual buyers & sellers, one market	Price of rice; wages in garment industry
Macroeconomics	Whole national economy	Bangladesh GDP growth; national inflation rate

A.2 Positive vs. Normative Economics

Positive Economics: Describes the world AS IT IS. Can be tested by data.
Example: "When rice price rises, quantity demanded falls."

Normative Economics: States what SHOULD happen. It is an opinion/value judgment.
Example: "The government SHOULD subsidize rice for poor families."

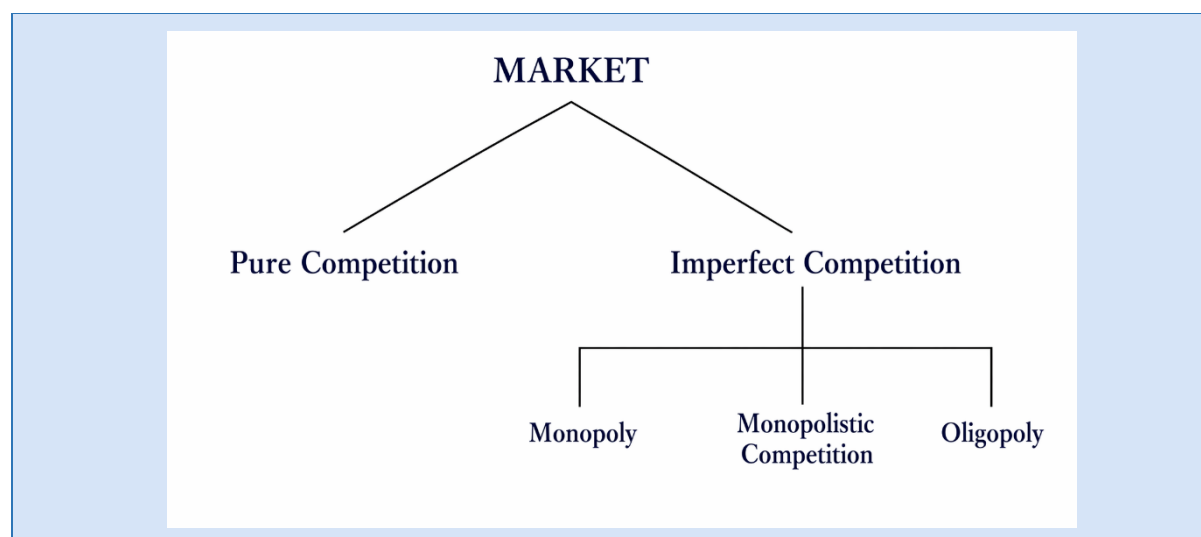
Type	Key Word	Can Be Tested?	Example
Positive	"Is" / "Are"	YES — by data	"Bangladesh exports more garments when price rises"
Normative	"Should" / "Ought"	NO — it is opinion	"Workers should be paid higher wages"

A.3 Scarcity and Choice

Scarcity = Unlimited Wants + Limited Resources.
This is the CENTRAL PROBLEM of economics.
Because resources are scarce, society must make CHOICES.
Every choice involves an OPPORTUNITY COST (what you give up).

SECTION B: Market Structures — Full Tree

Your notes have a table. The class lecture also showed this as a tree diagram:



Market Type	Sellers	Price Control	Bangladesh Example
Pure Competition	Many (infinite)	None — price taker	Rice market, vegetable bazaar
Monopolistic Competition	Many	Some (small)	Restaurants in Dhaka
Oligopoly	Few (2–5)	Significant	Grameenphone, Robi, Banglalink
Monopoly	One	Full (price maker)	DESA electricity, Dhaka

SECTION C: Supply Equation — Mathematical Form

Your existing notes explain the demand equation ($Q_d = a - bP$) but NOT the supply equation. From your class images:

Supply Function:

$Q_s = f(P)$ Quantity Supplied is a function of Price

$Q_s = c + dP$ Linear supply equation

Where:

c = quantity supplied when price is 0 (can be negative — seller needs min price)

d = how much Q_s rises per 1-unit rise in P (positive — upward slope)

The PLUS sign confirms the DIRECT (upward) relationship

Both together:

$Q_d = a - bP$ (Demand — negative slope — downward)

$Q_s = c + dP$ (Supply — positive slope — upward)

SECTION D: Finding Equilibrium Algebraically

Your existing notes explain equilibrium using graphs and words. From Images 6 & 7, the class showed the full algebraic method:

D.1 General Formula

At Equilibrium: $Q_d = Q_s$

Given:

$Q_d = a - bP$ and $Q_s = c + dP$

Step 1: Set them equal

$a - bP = c + dP$

Step 2: Rearrange (bring P terms together)

$a - c = bP + dP$

$a - c = P(b + d)$

Step 3: Solve for Equilibrium Price (P^*)

$P^* = (a - c) / (b + d)$

Step 4: Find Equilibrium Quantity (Q^*)

$Q^* = a - b \times P^*$ (substitute P^* into demand equation)

$Q^* = (ad + bc) / (b + d)$ [simplified form]

D.2 Worked Example

Given: $Q_d = 20 - 3P$ and $Q_s = 10 + 2P$ (from your class image)

Step 1: Set $Q_d = Q_s$

$$20 - 3P = 10 + 2P$$

Step 2: Rearrange

$$20 - 10 = 2P + 3P$$

$$10 = 5P$$

Step 3: Solve for P^*

$$P^* = 10 / 5 = 2$$

Step 4: Find Q^* using demand equation

$$Q^* = 20 - 3(2)$$

$$Q^* = 20 - 6 = 14$$

ANSWER:

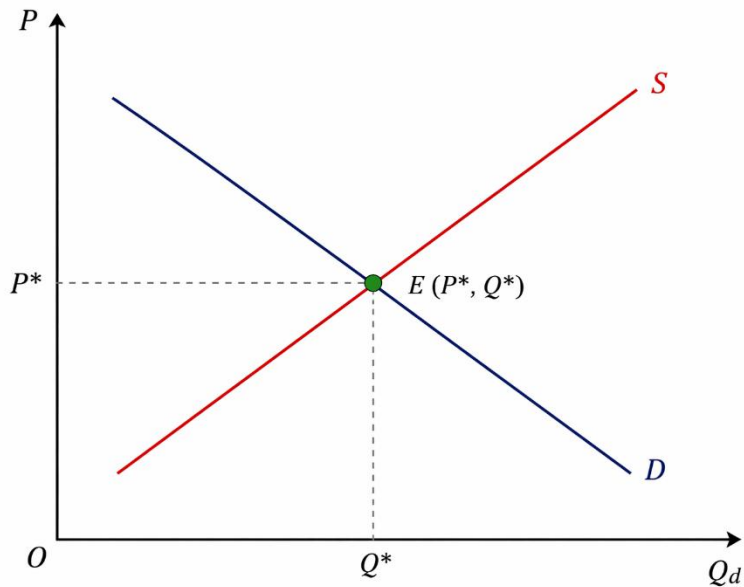
Equilibrium Price $P^* = 2$

Equilibrium Quantity $Q^* = 14$

Price (P)	$Q_d = 20 - 3P$	$Q_s = 10 + 2P$	Situation
1	17	12	$Q_d > Q_s \rightarrow$ Shortage
2	14	14	$Q_d = Q_s \rightarrow$ EQUILIBRIUM
3	11	16	$Q_s > Q_d \rightarrow$ Surplus

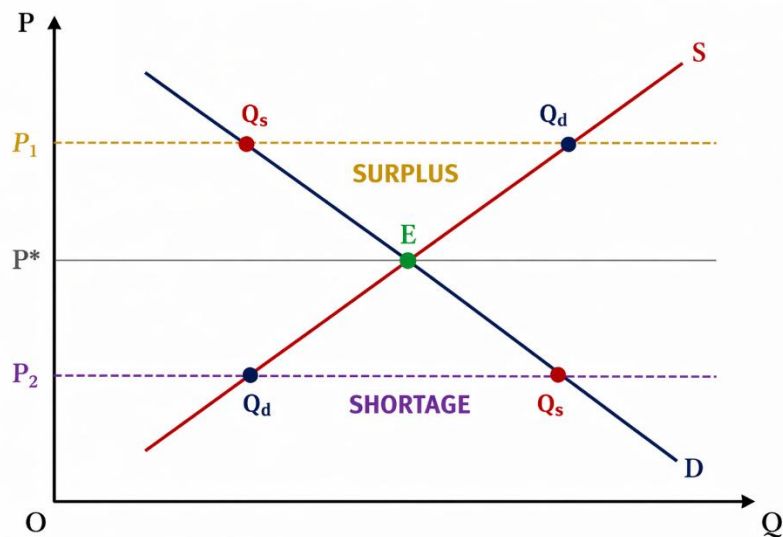
D.3 Graph: Equilibrium Point

Supply curve (S) and Demand curve (D) cross at equilibrium point E:



At E : $Q_d = Q_s = Q^*$. Price is P^* . Market is balanced — no pressure to change.

D.4 Graph: Surplus and Shortage Positions



Price Level	Situation	Who has problem?	Market Response
Above P^* (P_1)	SURPLUS — $Q_s > Q_d$	Sellers cannot sell all goods	Sellers cut price → returns to P^*
At P^*	EQUILIBRIUM — $Q_d = Q_s$	No one — market is balanced	Price stays stable

Below P^* (P_2)	SHORTAGE — $Q_d > Q_s$	Buyers cannot get enough	Sellers raise price → returns to P^*
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D.5 Consumer Surplus & Producer Surplus

Consumer Surplus: The benefit buyers receive.

Area = ABOVE the price line P^* and BELOW the demand curve D .

Triangle on top side of P^* line.

Producer Surplus: The benefit sellers receive.

Area = BELOW the price line P^* and ABOVE the supply curve S .

Triangle below P^* line.

Total Surplus = Consumer Surplus + Producer Surplus

Measures the total benefit to society from trade.

SECTION E: Demand Curve — Slope Calculation

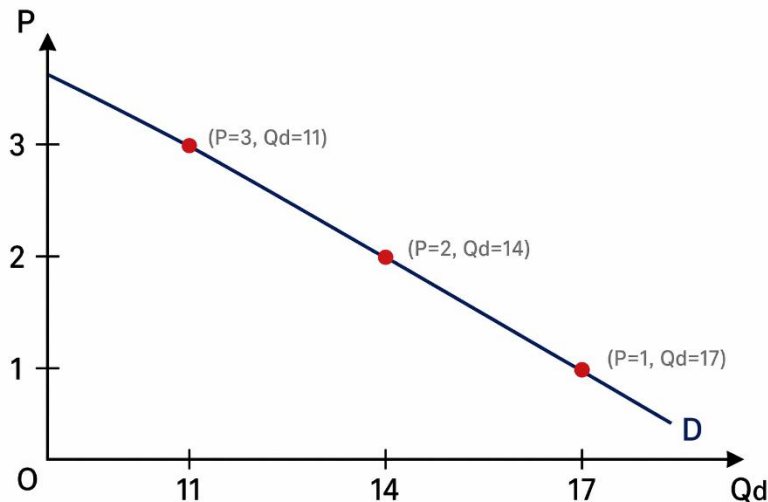
From Image 4 — your class worked this with actual points. This method is NOT in your original notes.

E.1 Points on the Demand Curve

Given equation: $Q_d = 20 - 3P$

Price (P)	Calculation	Quantity Demanded (Q_d)	Point on graph
$P = 3$	$20 - 3(3) = 11$	11 units	$(P=3, Q_d=11)$
$P = 2$	$20 - 3(2) = 14$	14 units	$(P=2, Q_d=14)$
$P = 1$	$20 - 3(1) = 17$	17 units	$(P=1, Q_d=17)$

E.2 Graph: Demand Curve with Three Points



E.3 Slope Calculation

Slope of Demand Curve

Slope of Demand Curve = Change in P / Change in Qd = $\Delta P / \Delta Qd$

Using points (P = 2, Qd = 14) and (P = 3, Qd = 11):

$$\Delta P = 2 - 3 = -1$$

$$\Delta Qd = 14 - 11 = 3$$

$$\text{Slope} = -1 / 3 = -1/3$$

Meaning:

For every 3-unit INCREASE in Qd, price FALLS by 1 unit.

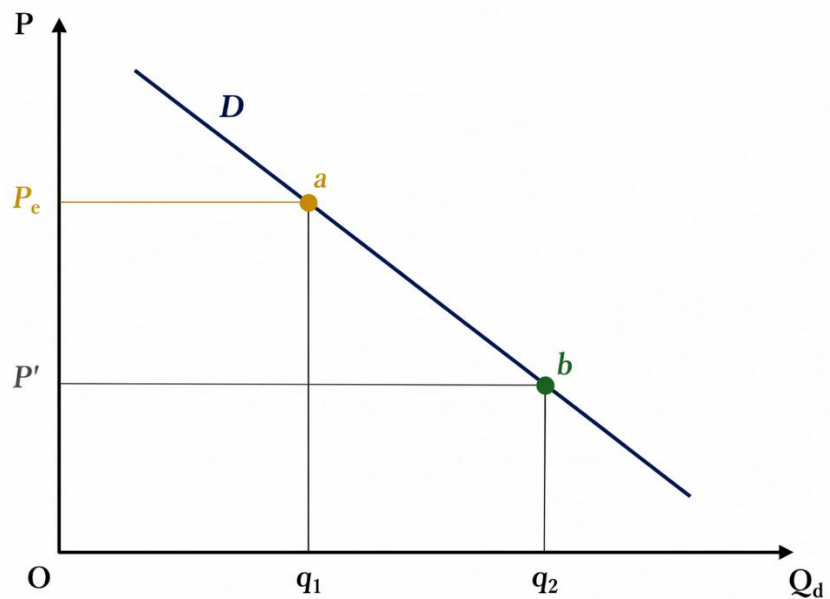
The slope is NEGATIVE → confirms the demand curve slopes DOWNWARD.

SECTION F: Movement vs. Shift — With Diagrams

Images 4 & 5 show two separate diagrams for this. Your notes explain it but without these specific diagrams.

F.1 Movement ALONG the Demand Curve

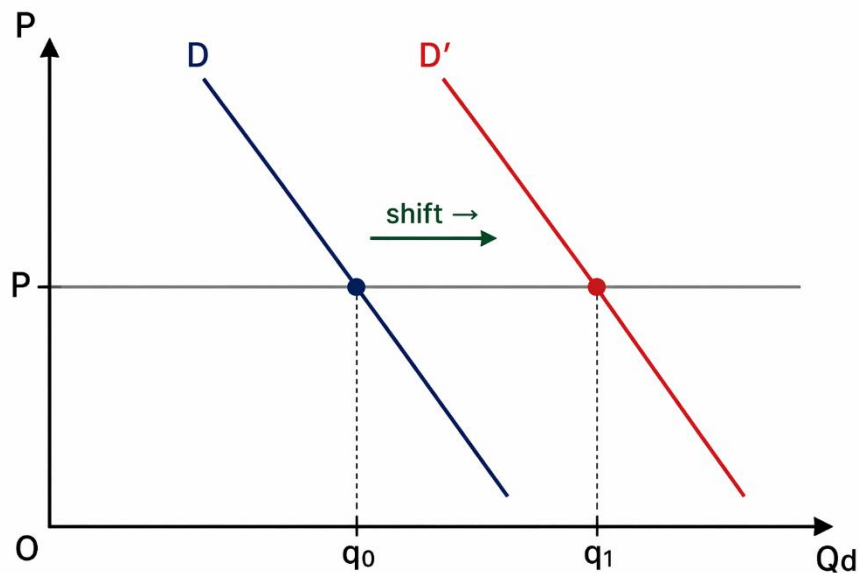
Caused by: PRICE of the same good changes. The curve stays in place — only the point moves.



Point moves from "a" to "b" on the SAME curve. Quantity changes from q_1 to q_2 . This is called: Change in Quantity Demanded.

F.2 Shift OF the Demand Curve

Caused by: Non-price factors (income, tastes, related goods, expectations, # buyers). The ENTIRE curve moves.



The whole curve moves from D to D' . At the SAME price P , quantity demanded increases from q_0 to q_1 . This is called: Change in Demand.

	Movement Along Curve	Shift of Curve
Cause	Price of the SAME good changes	Income, tastes, related goods, expectations, # buyers
What moves?	One point slides on same curve	Entire curve moves to new position
Technical name	Change in Quantity Demanded	Change in Demand
Graph effect	Move up or down same line	A new curve appears left or right

SECTION G: Price Elasticity of Demand (PED)

Images 9 & 10 cover elasticity. This topic is completely missing from your existing notes.

G.1 What Is Elasticity?

Price Elasticity of Demand (E_p or E_d) measures how SENSITIVE quantity demanded is to a change in price.

Question: If price rises by 10%, by how much does Q_d fall?
The answer is the elasticity value.

G.2 Formula

Price Elasticity of Demand (E_p):

$$E_p = \% \text{ Change in } Q_d / \% \text{ Change in } P$$

Expanded form:

$$E_p = (\text{Change in } Q_d / \text{Initial } Q_d) \div (\text{Change in } P / \text{Initial } P)$$

Point elasticity (math/calculus form):

$$E_p = (dQ / dP) \times (P / Q)$$

Note:

E_p is always NEGATIVE for demand (inverse relationship).
We report $|E_p|$ (absolute value) to compare magnitudes.

G.3 Worked Example 1 (from Image 9)

Given: When $P = 10$, $Q = 200$. When $P = 8$, $Q = 250$.

Step 1: Find changes

$$dP = 10 - 8 = 2$$

$$dQ = 200 - 250 = -50$$

Step 2: Apply formula

$$E_p = (dQ / dP) \times (P / Q)$$

$$E_p = (-50 / 2) \times (10 / 200)$$

$$E_p = -25 \times 0.05$$

$$E_p = -1.25$$

Interpretation:

$$|E_p| = 1.25 > 1 \rightarrow \text{Demand is ELASTIC}$$

A 1% rise in price leads to a 1.25% fall in quantity demanded.

G.4 Worked Example 2 (from Image 10)

Given: $P = 10$, $Q = 200$, then $P = 9$, $Q = 300$.

Case 1:

$$dP = 1, dQ = 100$$

$$E_p = (100 / 1) \times (10 / 200)$$

$$E_p = 100 \times 0.05 = 5$$

$$|E_p| = 5 > 1 \rightarrow \text{Highly ELASTIC}$$

Case 2 (Image 10 bottom):

P changes from 10 to 5, Q from 160 to 210

$$dP = 5, dQ = 50$$

$$E_p = (50 / 5) \times (5 / 210)$$

$$E_p = 10 \times 0.024 = 0.024$$

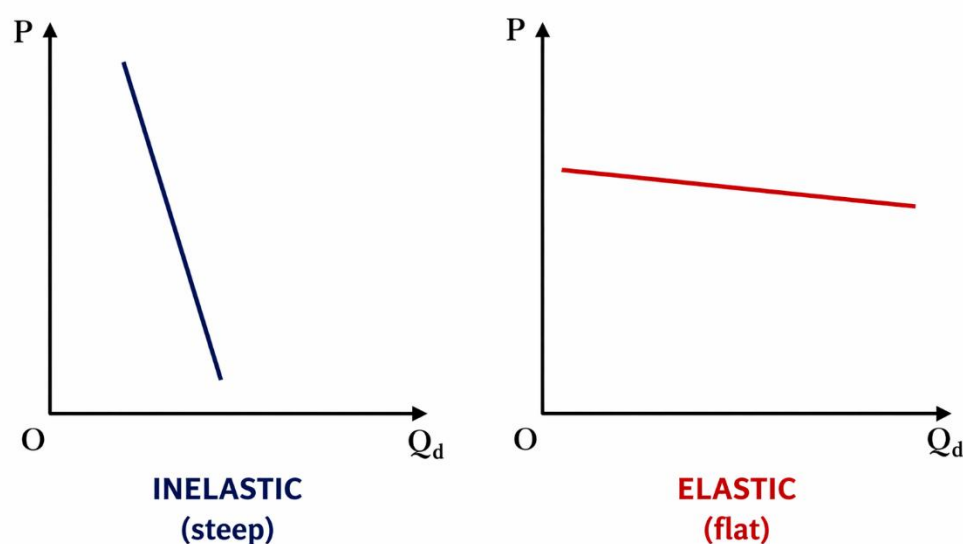
$$|E_p| = 0.024 < 1 \rightarrow \text{INELASTIC}$$

G.5 Elasticity Classification Table

Value of $ E_p $	Type	Meaning	Graph Shape
$ E_p > 1$	ELASTIC	Qd changes MORE than price change	Flat (gentle) curve

$ E_p = 1$	Unit Elastic	Qd changes by same % as price	Moderate slope
$ E_p < 1$	INELASTIC	Qd changes LESS than price change	Steep curve
$ E_p = 0$	Perfectly Inelastic	Qd does not change at all	Vertical line
$ E_p = \infty$	Perfectly Elastic	Any price rise $\rightarrow$ Qd drops to 0	Horizontal line

G.6 Graph: Elastic vs. Inelastic Demand



Left (steep/inelastic): Small change in Qd for a big change in P.

Right (flat/elastic): Large change in Qd for a small change in P.

SECTION H: Simultaneous Shift Scenarios

Image 8 shows specific scenarios not detailed in your existing notes:

Scenario	Demand Shift	Supply Shift	Price Result	Quantity Result
Consumer income $\uparrow$ + Input cost $\downarrow$	RIGHT (increase)	RIGHT (increase)	AMBIGUOUS	RISES for sure
Consumer income $\downarrow$ + Input cost $\downarrow$	LEFT (decrease)	RIGHT (increase)	FALLS for sure	AMBIGUOUS

Consumer income $\uparrow$ + Input cost $\uparrow$	RIGHT (increase)	LEFT (decrease)	RISES for sure	AMBIGUOUS
Consumer income $\downarrow$ + Input cost $\uparrow$	LEFT (decrease)	LEFT (decrease)	AMBIGUOUS	FALLS for sure

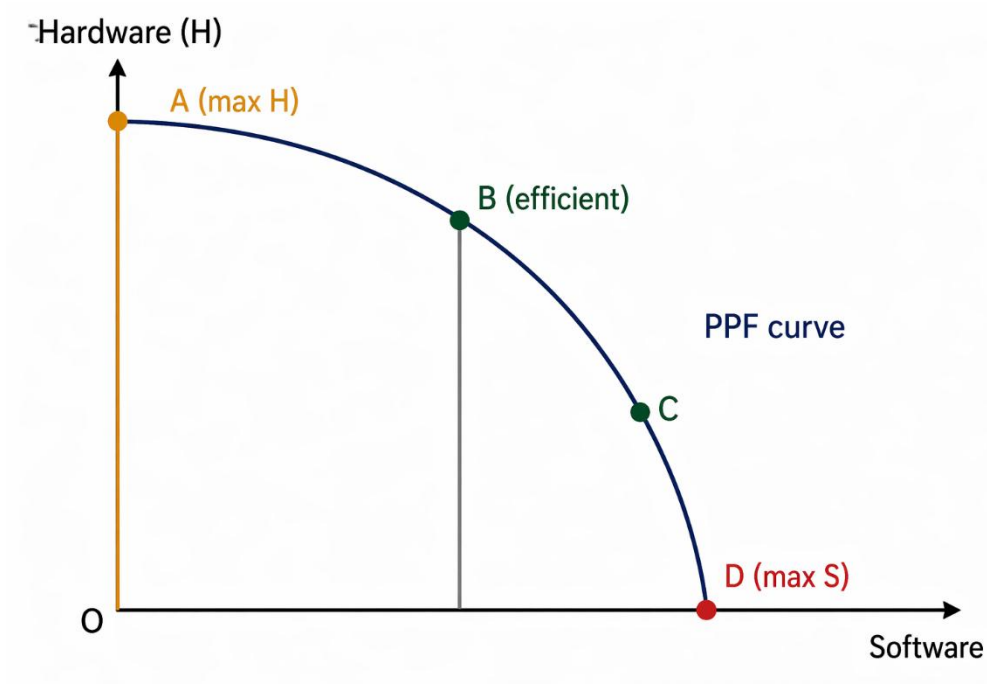
SECTION I: PPF — Production Possibility Frontier

Image 1 shows a PPF diagram. This is an economics model not in your Chapter 4 notes.

PPF = Production Possibility Frontier
Also called: PPC (Production Possibility Curve)

Definition: A curve showing ALL the maximum combinations of two goods that an economy can produce with its available resources and technology.

I.1 Graph: PPF Curve



Position	Meaning	Efficiency
ON the curve	Maximum output — using all resources	Efficient (best use of resources)
INSIDE the curve	Output below maximum — wasting resources	Inefficient
OUTSIDE the curve	Currently impossible — beyond capacity	Unattainable

I.2 Key Points about PPF

- The PPF is **CONCAVE** (bowed outward) due to increasing opportunity cost.
- To get more of one good, you must give up some of the other — this is opportunity cost.
- Economic growth **SHIFTS** the PPF outward (better technology or more resources).
- The PPF shows **SCARCITY** and **CHOICE** visually — you cannot have unlimited of both goods.

SECTION J: Economic Models & Circular Flow

Image 2 shows Economic Models branching into PPF and Circular Flow of GDP.

Economic Model: A simplified picture of reality used to understand how the economy works.

Two main models in Principles of Economics:

1. PPF (Production Possibility Frontier) — shows resource allocation and trade-offs
2. Circular Flow of GDP — shows money and goods flowing between households and firms

J.1 Circular Flow of GDP

$GDP = Y = C + I$ (simplified: no government, no foreign trade)

C = Consumption expenditure (households buy goods/services from firms)

I = Investment (firms buy capital goods)

S = Savings

Key equilibrium:

$I = S$ (Investment equals Savings in a simple closed economy)

Flow:

Households earn income (wages, rent) from firms via factor markets.

Households spend that income buying goods from firms via goods markets.

This circular flow keeps repeating — that is GDP.

QUICK REFERENCE — All New Formulas Added

Demand Equation:

$$Q_d = a - bP$$

Supply Equation:

$$Q_s = c + dP$$

Equilibrium Price:

$$P^* = (a - c) / (b + d)$$

Equilibrium Quantity:

$$Q^* = (ad + bc) / (b + d)$$

Demand Slope:

$$\Delta P / \Delta Q_d \text{ (always negative)}$$

Supply Slope:

$$\Delta P / \Delta Q_s \text{ (always positive)}$$

Price Elasticity:

$$E_p = (dQ/dP) \times (P/Q)$$

Elastic Demand:

$$|E_p| > 1$$

Inelastic Demand:

$$|E_p| < 1$$

GDP (simple):

$$Y = C + I$$

Savings = Investment:

$$I = S$$

